

59. Which one of the following is an example of in-sight learning?
- A. Salivation of a dog when a bell was rung in the absence of food, which was paired with the conditioned stimulus.
 - ☒ B. Use of boxes, after several trials, by a chimpanzee placed in an enclosed area to obtain bananas placed on the top of an enclosure.
 - C. Exhibition of modifications of natural behaviors of guide dogs, horses or zoo animals to desired behaviors.
 - D. Lack of response from a lady, which sat in a shop in which a strong deodorant was continuously released from a can.
60. Which one of the following is an example of territorial behaviour of animals?
- ☒ A. A male robin threatening an intruder
 - B. Fixed action of zebra fish to find a mate
 - C. Penguins huddling to reduce heat loss
 - D. Honeybees nesting in a tree
61. One of the following renowned Ethiopian biologists has developed a vaccine for the rinderpest disease.
- A. Professor Sebsebe Demissew
 - B. Professor Yalemtehay Mekonnen
 - C. Professor Beyene Petros
 - ☒ D. Professor Tilahun Yilma
62. A grade nine student has planned to study animal cells using a compound light microscope. She prepared a slide of stained human cheek cells for this purpose. Which one of the following activities should she first do?
- A. Adjust the iris diaphragm until the maximum brightness is coming.
 - ☒ B. Set up the microscope with the lowest power objective in place.
 - C. Using coarse focusing knob to rotate the nosepiece for a higher magnification lens.
 - D. Looking through the eyepiece lens to move the objectives down.
63. Which one of the following is true about plant cells? They have
- A. vacuole which controls all activities of the cell.
 - B. lysosomes which contain hydrolytic enzymes.
 - C. chloroplast which is filled with sap, liquids and minerals.
 - ☒ D. cell wall which gives support and strengthen to the cell.

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64. A plant cell is placed in a solution which has lower solute concentration than its contents. After a few minutes the cell will become
- ☒ A. turgid.
☐ B. flaccid.
☐ C. plasmolysed.
☐ D. burst.
65. Which one of the following is correct about human food digestion?
- ☐ A. The stomach physically breaks down the ingested food.
☐ B. Mouth use protease enzyme to breakdown proteins.
☐ C. Large intestine is the site of lipid hydrolysis.
☐ D. The small intestine is site of absorption of digested molecules.
66. Sickle cell anemia is caused by the replacement of glutamic acid by valine at a specific position in the beta chains of hemoglobin. This change is an example of which mutation?
- ☐ A. Addition
☒ B. Substitution
☐ C. Inversion
☐ D. Deletion
67. Stanley Miller conducted spark-discharge experiment by taking a mixture of gases that he thought to represent the primitive atmosphere of the Earth. All of the following were found in his gas mixture **EXCEPT**
- ☐ A. NH_3
☐ B. CH_4
☐ C. H_2
☒ D. CO_2
68. Which one of the following sentences explains the Darwin theory of evolution?
- ☐ A. Environment changes, creating a need for organism to change.
☒ B. There is a natural variation in features that are heritable.
☐ C. New features acquired in lifetime are passed to the offspring.
☐ D. There is development of new feature in order to survive.
69. Comparative anatomy provides evidence of evolution because it
- ☒ A. looks at structural similarities of organisms.
☐ B. has remains of animals and plants from the remote past.
☐ C. has an impression of animal even if the actual parts are missing.
☐ D. studies the development of the embryos of vertebrates.

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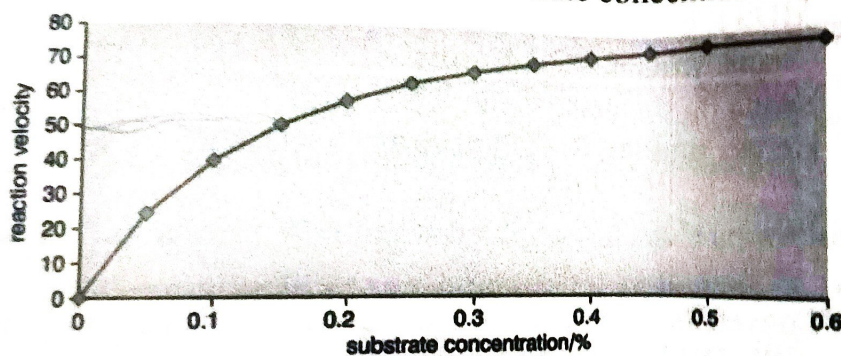
70. About 3 million years ago the Isthmus of Panama separated the shrimp population of the Caribbean Sea and created two populations of shrimps, one on either side of the isthmus. They still look remarkably similar, but they do not interbreed with each other. Which type of isolation is described by this scenario?

- A. Seasonal isolation
- B. Geographical isolation
- C. Behavioural isolation
- D. Temporal isolation

71. In a metabolic pathway, the end-product inhibition occurs when the

- A. last product of the pathway inhibits the last enzymatic reaction.
- B. first product of the pathway inhibits the last enzymatic reaction.
- C. first product of the pathway inhibits the third enzyme of the reaction.
- D. last product of the pathway inhibits the first enzymatic reaction.

72. The following graph shows the rate of reaction of catalase enzyme at a constant temperature and at different substrate concentration (in %).



From the above graph, we can conclude that catalase was functional with its minimum turnover rate from substrate concentration of

- A. 0.4 to 0.5.
- B. 0.1 to 0.2.
- C. 0.5 to 0.6.
- D. 0.3 to 0.4.

73. Which one of the following statements is correct about the description of the cell theory?

- A. Schleiden and Schwann stated that all cells come from pre-existing cells.
- B. Virchow stated that cell is the unit of structure, physiology and organization of living things.
- C. Modern cell theory states that cells contain hereditary information that passes from cell to cell.
- D. Spontaneous generation theory states that all cells have basically different chemical composition.

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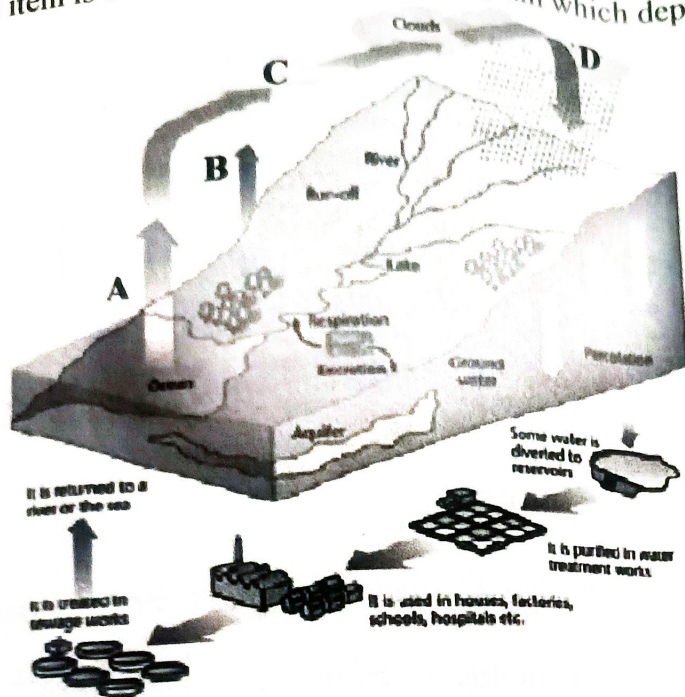
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74. A student measured the diameter of onion epidermal cells at a magnification of 100X using a field of view adjusted to 2mm by stage micrometer. What would be the diameter of individual cells if 8 cells fit into the field of view of the microscope at 100X magnification?
- A. 250µm
B. 200µm
C. 150µm
D. 300µm
75. Which of the following characteristics of cells is found only in prokaryotic cell?
- A. Division of labor within the cell
B. Continuous loop of DNA
C. 10 -100 µm long
D. Higher level of complexity
76. Which one of the following sentences is true about viruses?
- A. AIDS is caused by a DNA virus.
B. Corona virus is a DNA virus.
C. Swine flu is caused by an RNA virus.
D. Herpes simplex is a DNA virus.
77. The following steps are involved when HIV infects its host cell in the human body.
1. Conversion of viral RNA into DNA and its incorporation into cell's DNA.
 2. Viral DNA's transcription into viral RNA and production of proteins.
 3. Assemblage of new viral particles from RNA, proteins and Reverse Transcriptase
 4. The binding of gp120 of HIV to CD4 of a T-Lymphocyte
 5. Fusion of HIV with the plasma membrane of the host cell
 6. Release of its RNA and reverse transcriptase into the cell.
- The logical order of the steps listed above would be
- A. 4, 5, 6, 1, 2, 3.
B. 4, 2, 5, 6, 1, 3.
C. 4, 6, 5, 2, 1, 3.
D. 4, 1, 2, 5, 6, 3.
78. Which one of the following bacteria produces ammonium ion by forming a symbiotic association with peas and beans?
- A. Klebsiella
B. Azotobacter
C. Psuedomonas
D. Rhizobium

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79. This item is based on the following diagram which depicts the water cycle.



- In the above diagram, the letters A,B,C and D respectively represent
- evaporation, transpiration, condensation, precipitation.
 - evaporation, precipitation, transpiration, condensation.
 - evaporation, transpiration, precipitation, condensation.
 - evaporation, precipitation, condensation, transpiration.

80. Which one of the following comparisons is true about primary and secondary successions? Secondary succession has
- reduced species diversity.
 - reduced total biomass of the community.
 - has slower progress to reach to the climax.
 - more stable community.
81. Which one of the following sentences describes the tropical rainforest biome? It has
- poor and thin soil.
 - sparse - succulent plants.
 - frozen and rocky soil.
 - migrating animals.

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82. Fresh water biomes like streams and rivers are characterized by the presence of
A. corals, crabs and planktons.
B. algae, planktons and fish.
C. corals, kelps and angler fish.
D. angler fish, algae and crabs.
83. Which one of the following statements correctly describes intra-specific and inter-specific competitions? In intra-specific competition the
A. competition is less severe but in inter-specific competition the
B. competition is for a few requirements but in inter-specific competition
C. competing individuals have similar adaptation but in inter-specific
D. competition the adaptation is different.
competition the individuals belong to different species but in inter-specific
competition the individuals are of the same species.
84. Which one of the following stages of cell division is correctly described in meiosis II?
A. Prophase II - pairs of homologous chromosomes split up
B. Anaphase II - chromatids from each chromosome separate
C. Metaphase II - chromosomes pair and exchange segments
D. Telophase II - two haploid cells are produced
85. Which one of the following events is an example of a sex influenced trait?
A. Pattern baldness
B. Lactation
C. Ritinitis pigmentosa
D. Undescended testicles
86. Which of the following processes takes place when lactate is produced?
A. NAD and FAD are regenerated from reduced NAD and FAD.
B. ATP is produced through oxidative phosphorylation.
C. Protons accumulation created in the inter-membrane space.
D. Reduced NAD supplies hydrogen and becomes oxidized itself.
87. Which one of the following structures of the photosystem is correctly related with its function?
A. Chlorophyll a - a reaction center molecule
B. Antenna complex -light dependent reaction begins
C. Carotenoid - provides the hydrogen ions
D. Thylakoid - absorbs blue and red pigments

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88. What happens during the electron transport system of respiration? The four-carbon compound undergoes molecular transformation to
- A. generate oxaloacetate.
 - B. hydrogen atoms carried by FADH and NADH are released and split into protons and electrons.
 - C. citrate losses a carbon atom to form a five-carbon atom compound and CO_2 .
 - D. five carbon compound is further decarboxylated to form a four-carbon compound.
89. During light independent reactions of photosynthesis, glucose is synthesized through
- A. synthesis of reduced NADP that provides hydrogen ion.
 - B. six turns of the Calvin cycle to make one molecule of glucose.
 - C. release of hydrogen ions at the electron transport chain.
 - D. transmission of electrons to the inside of the thylakoid.
90. Which one of the following factors of photosynthesis is correctly matched with its effect?
- A. CO_2 concentration - limits the reactions by influencing rate of initial reaction with RuBP
 - B. Low temperature - limits the rate of reactions by reducing the number of electrons in the chlorophyll
 - C. Low Rubisco - limits the number of reactions in the light dependent reaction
 - D. Light intensity - limits the rate of ATP synthase enzyme action in light dependent reactions
91. Which one of the following monohybrid crosses can be a test cross?
- | | |
|---------------------------------|---------------------------------|
| A. $\text{Rr} \times \text{Rr}$ | C. $\text{RR} \times \text{rr}$ |
| B. $\text{RR} \times \text{Rr}$ | D. $\text{RR} \times \text{RR}$ |
92. Which of the following is different from the rest with regard to conservation of vegetation?
- A. Using local practices for conservation
 - B. Replanting of land with endemic species
 - C. Replanting indigenous tree species
 - D. Large scale introduction of exotic plants

93. In the nervous co-ordination system, which of the following statements explains the synapse? It is a
- chemical that transmits nerve impulses across the nervous system.
 - part of the skull that encloses brain for the purpose of protection.
 - junction between two neurons or a neuron and a muscle.
 - short term change in the electrical potential on the surface of cells.
94. Two students want to demonstrate a simple spinal reflex action using the knee jerk reflex. One of them sits down with one leg loosely crossed over the other. The other hits the crossed leg just below the knee cap gently but firmly using the edge of his hand. Which one of the following situations will happen during this demonstration? The
- impulse travels to the brain.
 - hit ligament stretches the muscle.
 - lower leg to jerk downwards and inwards.
 - impulse travels to quadriceps muscle from the brain.
95. Which of the following eye structures is correctly matched with its function?
- Pupil - serves as the eye's protective outer layer.
 - Iris - controls the amount of light entering the eye.
 - Cornea - provides nutrition to the retina.
 - Choroid - bends the light into the eye.
96. In the CAM photosynthetic pathway,
- malate is stored in the mesophyll cells during the day.
 - glucose is synthesized in the bundle sheath cell during night.
 - ATP and NADPH are produced during the night.
 - the C4 stage and the Calvin cycle are separated in time.
97. In Gram staining, if some bacteria retain the crystal violet after counter staining, then the bacteria is
- | | |
|--|---|
| ☒ A. Gram positive. | ☐ C. Gram intermediate. |
| ☐ B. Gram negative. | ☐ D. Gram variable. |
98. Which one of the following sentences explains the mechanism of bacterium to produce disease?
- Its hyphae secrete enzymes which digest substances in the tissue.
 - It enters into the living cells and disrupts the metabolic systems.
 - It attaches with white blood cells and destroy the immune system.
 - Its genetic material is incorporated into the cell for instruction.

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99. Which one of the following sentences explains the role of bacteria in genetic engineering? They are used to

- A. isolate genes from donor DNA.
- B. open up a plasmid.
- C. cut genes from donor DNA.
- ☒ D. contain a recombinant plasmid.

100. Which one of the following sentences describes a virus? It

- A. has differentiated organelles.
- B. is saprophytic.
- ☒ C. is acellular.
- D. undergoes respiration.

The End